Chapter 2

Probability and distributions

Exercises

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2.1 Discrete random variable

|||| Exercise 2.1 Discrete random variable

a) Let X be a stochastic variable. When running `stats.binom.pmf(4, 10, 0.6)` in Python the result is 0.1115:

```
print(stats.binom.pmf(4, 10, 0.6))
0.1115
```

What distribution is applied and what does 0.1115 represent?

|||| Solution

The binomial distribution with $n = 10$ and $p = 0.6$. The value 0.1115 is $P(X = 4)$: the probability of exactly 4 successes in 10 independent draws, each with success probability 60%.

b) Let X be the same stochastic variable as above. Given that $P(X \leq 4) = 0.1662$ and $P(X \leq 5) = 0.3669$, calculate $P(X \leq 5)$, $P(X < 5)$, $P(X > 4)$ and $P(X = 5)$.

|||| Solution

$$
\begin{aligned}
P(X \leq 5) &= 0.3669, \\
P(X < 5) &= P(X \leq 4) = 0.1662, \\
P(X > 4) &= 1 - P(X \leq 4) = 1 - 0.1662 = 0.8338, \\
P(X = 5) &= P(X \leq 5) - P(X \leq 4) = 0.3669 - 0.1662 = 0.2007.
\end{aligned}
$$

Or in Python:

```
print(stats.binom.cdf(5, 10, 0.6))
0.3669

print(stats.binom.cdf(4, 10, 0.6))
0.16629

print(1-stats.binom.cdf(4, 10, 0.6))
0.8338

print(stats.binom.pm    f(5, 10, 0.6))
0.2007
```

c) Let X be a stochastic variable. Running `stats.poisson.pmf(4, 3)` returns 0.168:

```
print(stats.poisson.pmf(4, 3))
0.168
```

What distribution is applied and what does 0.168 represent?

|||| Solution

The Poisson distribution with $\lambda = 3$. The value 0.168 is $P(X = 4)$: the probability of observing exactly 4 events per interval when the average rate is $\lambda = 3$ events per interval.

d) Let X be the same Poisson variable as above. Given that $P(X \leq 4) = 0.8153$ and $P(X \leq 5) = 0.9161$, calculate $P(X \leq 5)$, $P(X < 5)$, $P(X > 4)$ and $P(X = 5)$.

Solution

$$
\begin{aligned}
P(X \leq 5) &= 0.9161, \\
P(X < 5) = P(X \leq 4) &= 0.8153, \\
P(X > 4) = 1 - P(X \leq 4) &= 1 - 0.8153 = 0.1847, \\
P(X = 5) = P(X \leq 5) - P(X \leq 4) &= 0.9161 - 0.8153 = 0.1008.
\end{aligned}
$$

Or in Python:

```
print(stats.poisson.cdf(5, 3))
0.9161

print(stats.poisson.cdf(4, 3))
0.8153

print(1-stats.poisson.cdf(4, 3))
0.1847

print(stats.poisson.pmf(5, 3))
0.1008
```

2.2 Course passing proportions

Exercise 2.2 Course passing proportions

a) If a passing proportion for a course given repeatedly is assumed to be 0.80 on average, and there are 250 students taking the exam each time, what is the expected value μ and standard deviation σ for the number of students who do *not* pass the exam for a randomly selected course?

Solution

Let X be the number of students not passing. Then $X \sim B(n = 250,\ p = 0.20)$, and the mean and variance formulas for the binomial give:

$$\mu = np = 0.20 \times 250 = 50,$$
$$\sigma^2 = np(1-p) = 250 \times 0.20 \times 0.80 = 40, \quad \sigma = \sqrt{40} = 6.32.$$

So $\mu = 50$ and $\sigma = 6.32$.

Exam-style questions

b) A ferry company reports that, on average, 12% of its crossings are delayed due to weather. Over a season there are $n = 180$ crossings. Let Y denote the number of delayed crossings. The following code was run in Python:

```
print(stats.binom.mean(180, 0.12))
21.6

print(stats.binom.std(180, 0.12))
4.36
```

Which statement correctly interprets this output?

1) On average 21.6 crossings are delayed per season, with a standard deviation of about 4.36 crossings.
2) On average 21.6 crossings are delayed per season, with a standard deviation of about 4.36%.

3) The probability of a delay is 21.6%, with an uncertainty of 4.36%.
4) Exactly 21.6 crossings will be delayed every season.
5) On average 4.36 crossings are delayed per season, with a standard deviation of about 21.6 crossings.

Solution

Correct answer: **1)**. With $Y \sim B(180, 0.12)$: $\mu = np = 180 \times 0.12 = 21.6$ and $\sigma = \sqrt{np(1-p)} = \sqrt{180 \times 0.12 \times 0.88} = \sqrt{19.008} = 4.359$.

Both the mean and standard deviation are counts of crossings, not percentages (rules out 2) and 3)). The mean is an average, not a guaranteed outcome, so it need not be an integer and does not mean exactly 21.6 crossings are delayed every season (rules out 4)). Option 5) swaps the mean and standard deviation.

2.3 Notes in a box

|||| Exercise 2.3 Notes in a box

A box contains 6 notes:

On 1 of the notes there is the number 1
On 2 of the notes there is the number 2
On 2 of the notes there is the number 3
On 1 of the notes there is the number 4

Two notes are drawn at random from the box without replacement. X describes the number of notes with the number 4 among the 2 drawn.

a) Find the mean and variance of X, and $P(X = 0)$.

|||| Solution

X follows the hypergeometric distribution with $N = 6$, $a = 1$ (one "4"-note), and $n = 2$ draws. The mean and variance are:

$$\mu_X = n\frac{a}{N} = \frac{2}{6} = \frac{1}{3},$$
$$\sigma_X^2 = n\frac{a(N-a)}{N^2} \cdot \frac{N-n}{N-1} = 2 \cdot \frac{1 \times 5}{36} \cdot \frac{4}{5} = \frac{8}{36} = \frac{2}{9}.$$

The probability formula gives:

$$P(X = 0) = \frac{\binom{1}{0}\binom{5}{2}}{\binom{6}{2}} = \frac{10}{15} = \frac{2}{3}.$$

So $\mu_X = 1/3$, $\sigma_X^2 = 2/9$ and $P(X = 0) = 2/3$.

b) The 2 notes are now drawn *with* replacement. What is the probability that none of the 2 notes has the number 1 on it?

Solution

With replacement, each draw is independent with success probability $p = 1/6$. The binomial pdf gives:

$$P(X = 0) = \binom{2}{0}\left(\frac{1}{6}\right)^0\left(\frac{5}{6}\right)^2 = \frac{25}{36} = 0.694.$$

```
print(stats.binom.pmf(0, 2, 1/6))
```

Exam-style questions

c) A quality inspector examines a shipment of 15 circuit boards, 4 of which are faulty. She randomly selects 3 boards without replacement for testing. Let W denote the number of faulty boards among the 3 selected. What is the probability that *exactly one* of the three selected boards is faulty?

1) 0.484
2) 0.363
3) 0.145
4) 0.267
5) 0.430

Solution

Correct answer: **1)**. W follows the hypergeometric distribution with $N = 15$, $a = 4$, $n = 3$:

$$P(W = 1) = \frac{\binom{4}{1}\binom{11}{2}}{\binom{15}{3}} = \frac{4 \times 55}{455} = 0.484.$$

Option 2) is $P(W = 0) = \binom{11}{3}/\binom{15}{3} = 0.363$: no faulty boards, not exactly one. Option 3) is $P(W = 2) = \binom{4}{2}\binom{11}{1}/\binom{15}{3} = 0.145$: two faulty boards, not one. Option 4) is the naive proportion $a/N = 4/15 = 0.267$, which ignores the combinatorics of choosing 3 boards altogether. Option 5) wrongly treats the draws as independent (i.e. *with* replacement), giving the binomial value $\binom{3}{1}(4/15)^1(11/15)^2 = 0.430$ instead of the hypergeometric one.

d) Suppose instead the inspector samples 3 boards *with* replacement from the same batch of 15 boards (4 faulty). Running

```
print(stats.binom(3, 4/15).mean())
```

gives `0.8`. If she instead samples 3 boards *without* replacement (as in the previous question), will the expected number of faulty boards found be higher, lower, or the same as 0.8?

1) The same, because the hypergeometric mean $n \cdot a/N$ is always numerically equal to the binomial mean np when $p = a/N$ — only the variance differs between sampling with and without replacement.
2) Higher without replacement, since removing a faulty board increases the chance that the next board is also faulty.
3) Lower without replacement, since a board that has already been tested can never be found faulty again.
4) It cannot be determined without knowing the exact order in which the boards are drawn.
5) Higher without replacement, because variance is always higher when sampling without replacement.

|||| Solution

Correct answer: **1)**. For the hypergeometric distribution, $\mu = n\,a/N = 3 \times 4/15 = 0.8$, exactly matching the binomial mean $np = 3 \times (4/15) = 0.8$. Sampling without replacement changes the *variance* (via the finite population correction factor $\frac{N-n}{N-1}$, which makes the hypergeometric variance *smaller* than the binomial one, the opposite of what option 5) claims) but never the mean. Options 2) and 3) describe plausible-sounding but incorrect intuitions about how removal without replacement would shift the average.

2.4 Consumer survey

Exercise 2.4 Consumer survey

In a consumer survey performed by a newspaper, 20 different groceries were purchased in a grocery store. Discrepancies between the price on the sales slip and the shelf price were found in 6 of these 20 products.

a) At the same time a customer buys 3 random groceries (without replacement) within the group of 20 groceries in the store. What is the probability that no discrepancies occur for this customer?

Solution

Let X denote the number of discrepancies when buying 3 different products from the 20. Since products are bought without replacement, X follows the hypergeometric distribution with $N = 20$, $a = 6$, $n = 3$:

$$P(X = 0) = \frac{\binom{6}{0}\binom{14}{3}}{\binom{20}{3}} = \frac{1 \times 364}{1140} = 0.319.$$

The probability can be also computed using Python:

```
print(stats.hypergeom.pmf(0, M=20, n=6, N=3))
0.319
```

Hence the probability that none of the 3 products have a discrepancy is 0.319.

OBS: notice the difference in notation between the books definitions and the Python commands (also see the appendix to find a "notation translation").

b) Let X denote the number of discrepancies when purchasing 3 random groceries (without replacement) within the group of 20 groceries in the store. What is the mean and variance of X?

Solution

X follows the hypergeometric distribution (without replacement) with $N = 20$, $a = 6$, $n = 3$:

$$\mu_X = n \cdot \frac{a}{N} = 3 \cdot \frac{6}{20} = 0.90,$$
$$\sigma_X^2 = n \cdot \frac{a}{N}\left(1 - \frac{a}{N}\right)\frac{N-n}{N-1} = 3 \cdot \frac{6}{20} \cdot \frac{14}{20} \cdot \frac{17}{19} = 0.564.$$

So $\mu_X = 0.90$ and $\sigma_X^2 = 0.56$.

The values can also be computed with Python:

```
print(stats.hypergeom.mean(M=20, n=6, N=3))
0.90

print(stats.hypergeom.var(M=20, n=6, N=3))
0.564
```

Exam-style questions

c) A batch of 30 lightbulbs contains 5 defective ones. An inspector randomly selects 6 bulbs without replacement for testing. What is the probability that *at least one* of the 6 selected bulbs is defective?

1) 0.702
2) 0.298
3) 0.167
4) 1.000
5) 0.833

Solution

Correct answer: **1)**. Let X be the number of defective bulbs among the 6 selected, $X \sim H(N = 30, a = 5, n = 6)$. Using the complement rule:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{\binom{5}{0}\binom{25}{6}}{\binom{30}{6}} = 1 - \frac{177100}{593775} = 1 - 0.298 = 0.702.$$

Option 2) is $P(X = 0)$ itself — the answer to "no defective bulbs", not "at least one", i.e. forgetting to take the complement.

Option 3) is the naive proportion $5/30 = 0.167$, ignoring the combinatorics of selecting 6 bulbs.

Option 4) would only be correct if it were certain that a defective bulb must be drawn, which it is not.

2.5 Hay delivery quality

|||| Exercise 2.5 Hay delivery quality

A horse owner receives 20 bales of hay in a sealed plastic packaging. To control the hay, 3 bales are randomly selected and each checked for harmful fungal spores.

It is believed that among the 20 bales, 2 are infected with fungal spores. A random variable X describes the number of infected bales among the three selected.

a) Find the mean of X, the variance of X, and $P(X \geq 1)$.

|||| Solution

X follows the hypergeometric distribution (sampling without replacement) with $N = 20$, $a = 2$, $n = 3$:

$$\mu_X = n\frac{a}{N} = 3 \cdot \frac{2}{20} = 0.3,$$
$$\sigma_X^2 = n\frac{a}{N}\left(1 - \frac{a}{N}\right)\frac{N-n}{N-1} = 3 \cdot \frac{2}{20} \cdot \frac{18}{20} \cdot \frac{17}{19} = 0.2416.$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{\binom{18}{3}}{\binom{20}{3}} = 1 - \frac{816}{1140} = 0.2842.$$

So $\mu_X = 0.3$, $\sigma_X^2 = 0.242$ and $P(X \geq 1) = 0.2842$.

```
print(stats.hypergeom(M=20, n=2, N=3).mean(),
      stats.hypergeom(M=20, n=2, N=3).var(),
      1 - stats.hypergeom.pmf(0, M=20, n=2, N=3))
```

b) Another supplier advertises that no more than 1% of his bales are infected. The horse owner buys 10 bales and decides to buy from this supplier for the rest of the season if all 10 bales are error-free. Find the probability that all 10 purchased bales are error-free if (a) 1% of bales are infected (p_1), and (b) 10% of bales are infected (p_{10}).

Solution

Since the supplier has a very large stock, we use the binomial distribution (sampling effectively with replacement from a large population):

$$p_1 = P(X = 0 \mid p = 0.01) = 0.99^{10} = 0.9044,$$
$$p_{10} = P(X = 0 \mid p = 0.10) = 0.90^{10} = 0.3487.$$

```
print(stats.binom.pmf(0, 10, 0.01), stats.binom.pmf(0, 10,
    0.10))
```

Exam-style questions

c) A vineyard harvests 25 crates of grapes, 3 of which have been exposed to frost damage. A wine inspector randomly selects 4 crates to test for damage. Let V denote the number of frost-damaged crates among the 4 selected. Which of the following gives the correct mean and variance of V?

1) $\mu = 0.48, \sigma^2 = 0.370$
2) $\mu = 0.48, \sigma^2 = 0.4224$
3) $\mu = 0.12, \sigma^2 = 0.370$
4) $\mu = 0.48, \sigma^2 = 0.48$
5) $\mu = 3.00, \sigma^2 = 0.370$

Solution

Correct answer: **1)**. With $V \sim \text{Hypergeometric}(N = 25, a = 3, n = 4)$:

$$\mu_V = n\frac{a}{N} = 4 \times \frac{3}{25} = 0.48,$$
$$\sigma_V^2 = n\frac{a}{N}\left(1 - \frac{a}{N}\right)\frac{N-n}{N-1} = 4 \times 0.12 \times 0.88 \times \frac{21}{24} = 0.370.$$

Option 2) omits the finite population correction factor $\frac{N-n}{N-1} = \frac{21}{24}$, using the binomial-style variance $np(1-p) = 0.4224$ instead. Option 3) uses the naive proportion $a/N =$

0.12 as the mean, instead of $n \cdot a/N$. Option 4) wrongly assumes variance equals the mean (a Poisson-style mistake). Option 5) confuses $a = 3$ itself for the mean.

d) A large medical device manufacturer claims that no more than 0.5% of its syringes are defective. A hospital buys a box of 20 syringes and will switch suppliers unless all 20 are defect-free. If the manufacturer's claim is accurate ($p = 0.005$), `stats.binom.pmf(0, 20, 0.005)` returns 0.905. If the true defect rate is actually $p = 0.05$ (ten times higher than claimed), what is the probability that all 20 syringes are still defect-free, and what does comparing the two probabilities suggest?

1) $0.95^{20} = 0.358$; this large drop shows that the "all defect-free" check has decent power to catch a shift from a 0.5% to a 5% defect rate.
2) $0.95^{20} = 0.358$; since both probabilities are below 1, the test cannot distinguish between the two defect rates.
3) $0.995^{20} \times 10 = 9.05$; a nonsensical value greater than 1, so the probability must be computed a different way.
4) $0.05^{20} \approx 0$; there is essentially no chance of an all defect-free box if $p = 0.05$.
5) 0.905; the probability is unaffected by an increase in p, since only 20 syringes are tested regardless of the true rate.

|||| Solution

Correct answer: **1)**. $P(\text{all defect-free}) = (1-p)^{20}$. For $p = 0.05$: $0.95^{20} = 0.358$, notably lower than 0.905 for $p = 0.005$: a supplier secretly shipping 10 times the claimed defect rate would still often pass the all-defect-free check, but far less often than a supplier truly meeting the claim — so the check does carry some (imperfect) power to detect the higher defect rate. Option 4) mistakenly raises p itself to the 20th power instead of $(1-p)$. Option 5) incorrectly claims the probability does not depend on p.

```
print(stats.binom.pmf(0, 20, 0.05))
```

2.6 Newspaper consumer survey

|||| Exercise 2.6 Newspaper consumer survey

In a consumer survey performed by a newspaper, 20 different groceries were purchased in a grocery store. Discrepancies between the price on the sales slip and the shelf price were found in 6 of these 20 products.

a) Let X denote the number of discrepancies when purchasing 3 random (different) products within the group of 20. What is the mean and variance of X?

|||| Solution

X follows the hypergeometric distribution (without replacement) with $N = 20$, $a = 6$, $n = 3$:

$$\mu_X = n \cdot \frac{a}{N} = 3 \cdot \frac{6}{20} = 0.90,$$
$$\sigma_X^2 = n \cdot \frac{a}{N}\left(1 - \frac{a}{N}\right)\frac{N-n}{N-1} = 3 \cdot \frac{6}{20} \cdot \frac{14}{20} \cdot \frac{17}{19} = 0.564.$$

With Python the answer would look like this:

```
print(stats.hypergeom.mean(M=20, n=6, N=3))
0.90

print(stats.hypergeom.var(M=20, n=6, N=3))
0.564
```

So $\mu_X = 0.90$ and $\sigma_X^2 = 0.56$.

2.7 A fully automated production

|||| Exercise 2.7 A fully automated production

On a large fully automated production plant, items are pushed to a side band at random time points, from which they are automatically fed to a control unit. The production plant is set up so that the number of items sent to the control unit on average is 1.6 items per minute. Let X denote the number of items pushed to the side band in 1 minute. It is assumed that X follows a Poisson distribution.

a) What is the probability that more than 5 items arrive at the control unit in a given minute?

|||| Solution

With $\lambda = 1.6$:

$$P(X > 5) = 1 - P(X \leq 5) = 0.006.$$

where $P(X \leq 5)$ must be computed using the Poisson distribution:

$$P(X \leq 5) = \sum_{x=0}^{5} \frac{\lambda^x}{x!} e^{-\lambda}$$

The value can be computed in Python (or by evaluating the sum directly).

To calculate the final answer directly we calculate in Python:

```
print(1 - stats.poisson.cdf(5, 1.6))
0.006
```

About 0.6% of minutes will have more than 5 items.

b) What is the probability that no more than 8 items arrive at the control unit within a 5-minute period?

Solution

Over 5 minutes the rate scales to $\lambda_{5\text{min}} = 5 \times 1.6 = 8$:

$$P(X \leq 8) = 0.593.$$

The value can be computed in Python:

```
print(stats.poisson.cdf(8, 8))
0.593
```

Exam-style questions

c) A monitoring system logs server errors at an average rate of $\lambda = 2.3$ errors per hour, assuming errors happen at random times.
Which of the following expressions gives, the probability that *at least* 4 errors occur wihtin a given hour?

1) `1 - stats.poisson.cdf(3, 2.3)`
2) `1 - stats.poisson.cdf(4, 2.3)`
3) `stats.poisson.cdf(4, 2.3)`
4) `stats.poisson.pmf(4, 2.3)`
5) `stats.poisson.cdf(3, 2.3)`

Solution

Correct answer: **1)**. $P(X \geq 4) = 1 - P(X \leq 3) = 1 -$ `stats.poisson.cdf(3, 2.3)`.

Option 2) is the given expression for $P(X > 4)$, one boundary off from what is asked.

Option 3) computes $P(X \leq 4)$.

Option 4) computes only $P(X = 4)$.

Option 5) computes $P(X \leq 3)$, forgetting to take the complement.

d) Continuing the same monitoring system ($\lambda = 2.3$ errors per hour), what is the probability that *no* errors occur during a **3-hour** shift?

1) $e^{-6.9} = 0.00101$
2) $e^{-2.3} = 0.1003$
3) $3e^{-2.3} = 0.301$
4) $1 - e^{-6.9} = 0.999$
5) $1/6.9 = 0.145$

|||| Solution

Correct answer: **1)**. Over a 3-hour shift the rate scales to $\lambda_{3\text{h}} = 3 \times 2.3 = 6.9$, so $P(X = 0) = e^{-\lambda_{3\text{h}}} = e^{-6.9} = 0.00101$.

Option 2) forgets to scale λ up to the 3-hour window and uses the 1-hour rate instead.

Option 3) makes the equally common (but equally wrong) mistake of scaling the *probability* by 3 rather than scaling λ before exponentiating.

Option 4) computes the complement, $P(X \geq 1)$, instead of $P(X = 0)$.

Option 5) is not based on the Poisson pmf at all.

2.8 Call center staff

Exercise 2.8 Call center staff

The staffing for answering calls in a company is based on 180 phone calls per hour randomly distributed. If there are 20 calls or more in a period of 5 minutes, the capacity is exceeded and there will be an unwanted waiting time. Hence the capacity is 19 calls per 5 minutes.

a) What is the probability that the capacity is exceeded in a random period of 5 minutes?

Solution

The 5-minute mean is $\lambda_{5\text{min}} = 180/12 = 15$. Let $X \sim \text{Po}(15)$:

$$P(X \geq 20) = 1 - P(X \leq 19) = 0.125.$$

Here $P(X \leq 19)$ must be evaluated as a rather long sum using the Poisson distribution. The value can be computed in Python:

```
print(1 - stats.poisson.cdf(19, 15))
0.125
```

There is a 12.5% chance the capacity is exceeded in a random 5-minute period.

b) If the probability of handling all calls without waiting time must be at least 99% for a randomly selected 5-minute period, how large should the capacity per 5 minutes be?

Solution

We need the smallest c (capacity) such that $P(X \leq c) \geq 0.99$ for $X \sim \text{Po}(15)$. We expect that the capacity must be greater than 19 and start by computing $P(X \leq c)$ for different values of c, that are larger than 19. Evaluating for $c = 22, 23, \ldots, 26$:

c	22	23	24	25	26
$P(X \leq c)$	0.9615	0.9762	0.9857	0.9915	0.9950

Since we require $P(X \leq c) \geq 0.99$, we see that c must be 25 or higher.

The values can be computed in Python:

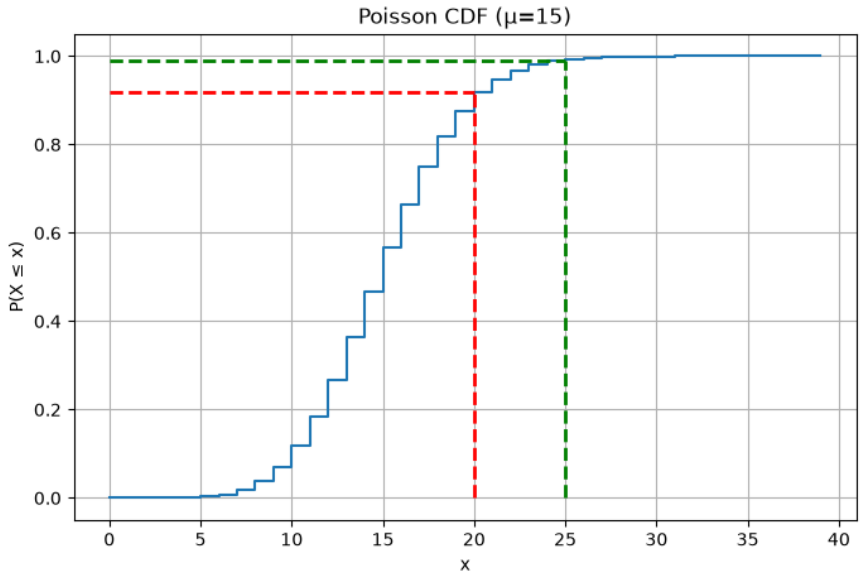
```
for c in range(22, 27):
    print(c, stats.poisson.cdf(c, mu=15))
22 0.9615
23 0.9762
24 0.9857
25 0.9915
26 0.9950
```

Alternatively we could use the ".ppf" function (the inverse of ".cdf"):

```
print(stats.poisson.ppf(q=0.99, mu=15))
25.0
```

The capacity must be at least 25 calls per 5 minutes.

Visualizing the calculated probabilities and quantiles:



Exam-style questions

c) A different company's helpdesk handles 240 calls per hour, with calls occuring at random times. If 12 or more calls arrive within a given 2-minute window, the helpdesk is considered overloaded. Let X denote the number of calls in a random 2-minute period. Which of the following Python expressions correctly computes the probability that the helpdesk is overloaded?

1) `1 - stats.poisson.cdf(11, 8)`
2) `1 - stats.poisson.cdf(12, 8)`
3) `stats.poisson.cdf(12, 8)`
4) `1 - stats.poisson.cdf(11, 4)`
5) `stats.poisson.pmf(12, 8)`

|||| Solution

Correct answer: **1)**. First convert the rate to the 2-minute window: $\lambda = 240 \text{ calls/hour} \times \frac{2}{60} \text{ hour} = 8$. "Overloaded" means $X \geq 12$, so $P(X \geq 12) = 1 - P(X \leq 11) =$ `1 - stats.poisson.cdf(11, 8)`.

Option 2) is off by one, computing $P(X > 12) = P(X \geq 13)$ instead.

Option 3) computes $P(X \leq 12)$, the wrong direction entirely.

Option 4) uses the correct boundary but the wrong rate $\lambda = 240/60 = 4$, forgetting to multiply by the 2-minute window length.

Option 5) computes only the single point probability $P(X = 12)$.

2.9 Continuous random variable

|||| Exercise 2.9 Continuous random variable

a) The following Python expressions evaluate to three different probabilities:

```
print(stats.norm.cdf(2))
print(stats.norm.cdf(2, 1, 1))
print(stats.norm.cdf(2, 1, 2))
```

Specify which distributions are used and explain the resulting probabilities.

|||| Solution

The three calls all use the normal cumulative distribution function (CDF). They return $P(X \leq 2)$ for:

- $X \sim N(\mu = 0,\ \sigma^2 = 1)$: $P(X \leq 2) = 0.9772$,
- $X \sim N(\mu = 1,\ \sigma^2 = 1)$: $P(X \leq 2) = 0.8413$,
- $X \sim N(\mu = 1,\ \sigma^2 = 4)$: $P(X \leq 2) = 0.6915$.

For the standard normal, $P(Z \leq 2) \approx 0.977$ represents the shaded area to the left of $z = 2$ under the bell curve.

b) What is the result of `stats.norm.ppf(stats.norm.cdf(2))`?

|||| Solution

`ppf` (percent-point function) and `cdf` are inverses of each other, so the result is 2.

c) The following expressions return three quantiles:

```
print(stats.norm.ppf(0.975))
print(stats.norm.ppf(0.975, 1, 1))
print(stats.norm.ppf(0.975, 1, 2))
```

State what each number represents.

Solution

Each number is the 97.5% percentile (i.e. the value x such that $P(X \leq x) = 0.975$) for:

- $N(\mu = 0,\ \sigma^2 = 1)$: $z_{0.975} = 1.960$,
- $N(\mu = 1,\ \sigma^2 = 1)$: 2.960 (shifted right by 1),
- $N(\mu = 1,\ \sigma^2 = 4)$: 4.920 (shifted right by 1 and stretched by 2).

Exam-style questions

d) Let $Y \sim N(10, 3^2)$. Given that `stats.norm.ppf(0.90)` $= 1.282$ (the 90th percentile of the standard normal), which of the following correctly gives the 90th percentile of Y?

1) $10 + 3 \times 1.282 = 13.85$
2) 1.282
3) $10 \times 1.282 = 12.82$
4) $(10 + 1.282) \times 3 = 33.85$
5) $10 - 3 \times 1.282 = 6.15$

Solution

Correct answer: **1)**. Quantiles of $N(\mu, \sigma^2)$ are obtained from standard-normal quantiles via $x = \mu + \sigma z$, so the 90th percentile of Y is $10 + 3 \times 1.282 = 13.85$. Note: This uses Theorem 2.43 ($Z = (X - \mu)/\sigma$), solved for X to get $x = \mu + \sigma z$.

```
print(stats.norm.ppf(0.90, 10, 3))
13.85
```

2.10 The normal pdf

Exercise 2.10 The normal pdf

a) Which of the following statements regarding the probability density function of the normal distribution $N(1, 2^2)$ is false?

1. The total area under the curve is equal to 1.0
2. The mean is equal to 1^2
3. The variance is equal to 2
4. The curve is symmetric about the mean
5. The two tails of the curve extend indefinitely

Solution

1. True — the total area under any pdf is 1.
2. True — the mean is 1, and indeed $1^2 = 1$.
3. **False** — for $N(1, 2^2)$ the variance is $2^2 = 4$, not 2.
4. True — the normal distribution is symmetric about its mean.
5. True — the normal density is defined on $(-\infty, \infty)$.

The correct (false) answer is **3**.

Let X be normally distributed with mean 24 and variance 16.

b) Calculate the following probabilities (using Python):
   - $P(X \leq 20)$
   - $P(X > 29.5)$
   - $P(X = 23.8)$

Solution

$X \sim N(24,\ 4^2)$.

```
print(stats.norm.cdf(20, 24, 4))
0.1587

print(1 - stats.norm.cdf(29.5, 24, 4))
0.0845
```

Since X is a continuous random variable, $P(X = 23.8) = 0$.

Exam-style questions

c) For $X \sim N(3, 5^2)$, which of the following statements is false?

1) The variance of X is 25.
2) The median of X equals its mean, 3.
3) Approximately 95% of the probability mass lies within 2 standard deviations of the mean, i.e. in the interval $[-7, 13]$.
4) The distribution of X is fairly left-skewed, since $\sigma = 5$ is fairly large.
5) $P(X < 3) = P(X > 3) = 0.5$.

Solution

Correct answer: **4)**. Every normal distribution is perfectly symmetric and therefore not left-skewed.

The other statements are all true: $\sigma^2 = 5^2 = 25$; the normal distribution's median always equals its mean; about 95% of the mass lies within $\mu \pm 2\sigma = 3 \pm 10 = [-7, 13]$; and by symmetry $P(X < 3) = P(X > 3) = 0.5$.

d) Let Y be normally distributed with mean 52 and standard deviation 6. Which of the following Python expressions correctly computes $P(45 < Y \leq 60)$?

1) `stats.norm.cdf(60, 52, 6) + stats.norm.cdf(45, 52, 6)`

2) `stats.norm.cdf(60, 52, 6) - stats.norm.cdf(45, 52, 6)`

3) `stats.norm.cdf(45, 52, 6) - stats.norm.cdf(60, 52, 6)`

4) `stats.norm.pdf(60, 52, 6) - stats.norm.pdf(45, 52, 6)`

5) `1 - stats.norm.cdf(60, 52, 6) - stats.norm.cdf(45, 52, 6)`

|||| Solution

Correct answer: **2)**. For any continuous distribution, $P(a < Y \leq b) = P(Y \leq b) - P(Y \leq a) = F(b) - F(a)$, so here `cdf(60,...)` $-$ `cdf(45,...)` ≈ 0.787.

```
print(stats.norm.cdf(60, 52, 6) - stats.norm.cdf(45, 52, 6))
0.787
```

2.11 Computer chip control

|||| Exercise 2.11 Computer chip control

A machine for checking computer chips uses on average 65 milliseconds per check with a standard deviation of 4 milliseconds. A newer machine, potentially to be bought, uses on average 54 milliseconds per check with a standard deviation of 3 milliseconds. Check times can be assumed normally distributed and independent.

a) What is the probability that the time savings per check using the new machine is less than 10 milliseconds?

|||| Solution

Let $X_{\text{old}} \sim N(65, 4^2)$ and $X_{\text{new}} \sim N(54, 3^2)$. The time saving $U = X_{\text{old}} - X_{\text{new}}$ follows a normal distribution with:

$$\begin{aligned} \mathrm{E}(U) &= 65 - 54 = 11, \\ \mathrm{V}(U) &= 4^2 + 3^2 = 25, \\ U &\sim N(11,\ 5^2). \end{aligned}$$

We need to find:

$$P(U < 10) = F(10)$$

The value is computed in Python:

```
print(stats.norm.cdf(10, 11, 5))
0.4207
```

b) What is the mean μ and standard deviation σ for the total time to check 100 chips on the new machine?

Solution

Let $U = \sum_{i=1}^{100} X_i$ with each $X_i \sim N(54, 3^2)$. U will then follow a normal distribution ($N(\mu, \sigma^2)$) with:

$$\mu = \mathrm{E}(U) = \sum_{i=1}^{100} 54 = 5400 \text{ ms},$$
$$\sigma^2 = \mathrm{V}(U) = \sum_{i=1}^{100} 9 = 900, \quad \sigma = 30 \text{ ms}.$$

Hence $\mu = 5400$ ms and $\sigma = 30$ ms.

Note: This uses Theorem 2.56 (mean and variance of linear combinations) with all weights $a = 1$, so it reduces to $E(U) = \sum E(X_i)$ and $V(U) = \sum V(X_i)$. Since each $X_i \sim N(54, 3^2)$, this gives $E(U) = 10054 = 5400$ ms and $V(U) = 1009 = 900$, so $\sigma = 30$ ms.

Exam-style questions

c) A courier company's Route A delivery time is $X_A \sim N(40, 6^2)$ minutes and Route B's is $X_B \sim N(35, 4^2)$ minutes, independent of each other. Let $D = X_A - X_B$ be the difference in delivery times. What is the variance of D?

1) $\sigma_D^2 = (6+4)^2 = 100$
2) $\sigma_D^2 = 6^2 - 4^2 = 20$
3) $\sigma_D^2 = (6-4)^2 = 4$
4) $\sigma_D^2 = 6^2 \times 4^2 = 576$
5) $\sigma_D^2 = 6^2 + 4^2 = 52$

Solution

Correct answer: **5)**. For independent random variables, variances always *add*, even when the variables themselves are subtracted: $\mathrm{V}(X_A - X_B) = \mathrm{V}(X_A) + \mathrm{V}(X_B) = 6^2 + 4^2 = 52$.

d) Continuing the courier company, Route B's delivery time is $X_B \sim N(35, 4^2)$ minutes, independent across deliveries. Which of the following correctly calculates the probability that the total time for 50 independent Route B deliveries exceeds 1770 minutes?

1)

```
print(stats.norm.cdf(1770, 1750, 16))
0.894
```

2)

```
print(stats.norm.cdf(1770, 1750, 800))
0.510
```

3)

```
print(1 - stats.norm.cdf(1770, 1750, np.sqrt(800)))
0.240
```

4)

```
print(stats.norm.cdf(1770, 1750, 200))
0.540
```

5)

```
print(1 - stats.norm.cdf(1770, 1770, 16))
0.500
```

|||| Solution

Let $U = X_1 + X_2 + \cdots + X_{50}$, the total delivery time for the 50 Route B deliveries. Each $X_i \sim N(35, 4^2)$, independently.

U is a linear combination of the X_i with every weight $a_i = 1$, so Theorem 2.56 gives:

$$E(U) = a_1 E(X_1) + \cdots + a_{50} E(X_{50}) = \sum_{i=1}^{50} E(X_i) = 50 \times 35 = 1750,$$

$$V(U) = a_1^2 V(X_1) + \cdots + a_{50}^2 V(X_{50}) = \sum_{i=1}^{50} V(X_i) = 50 \times 4^2 = 800.$$

So $\mu_U = 1750 \approx 28.28$ and $\sigma_U = \sqrt{800}$.

A sum of independent normal random variables is also normaly distributed, so

$$U \sim N(1750, 800).$$

We need to calculate the probability; $P(U > 1770) = 1 - P(U \leq 1770) = 1 - F(1770)$:

```
print(1 - stats.norm.cdf(1770, 1750, np.sqrt(800)))
0.240
```

Correct answer: **3)**.

2.12 Concrete items

Exercise 2.12 Concrete items

A manufacturer of concrete items knows that the length L of his items is reasonably normally distributed with $\mu_L = 3000$ mm and $\sigma_L = 3$ mm. The requirement is that the length must satisfy 2993 mm $\leq L \leq$ 3007 mm.

a) What is the expected error rate in the manufacturing?

Solution

The expected error rate is:

$$P(L < 2993) + P(L > 3007) = 2 \cdot P(L < 2993)$$

The value is computed with Python:

```
print(2*stats.norm.cdf(2993,3000,3))
0.0196
```

About 2% of items will not meet the specification.

b) The concrete items are supported by two beams - one at each end. For the items to be supported correctly, the distance between the two beams (called L_{beam}), must be approximately 100 mm shorter than the length of the concrete item itself (still called L). The specific requirement is:

$$90 \text{ mm} < L - L_{\text{beam}} < 110 \text{ mm}$$

The distance between beams, L_{beam}, is normally distributed with mean $\mu_{\text{beam}} = 2900$ mm. The question is now how precise this distance between beams (L_{beam}) has to be, i.e., how *small* should the standard deviation of L_{beam} be, in order for the requirement above to be fulfilled *most of the time*? Specifically calculate the standard deviation σ_{beam} if the requirement must be fulfilled in 99% of cases.
(Quantile from standard normal distribution: $z_{0.995} = 2.576$).

|||| Solution

Step 1: Identify the two random variables.

Let $L \sim N(3000, 3^2)$ be the length of the concrete item, and $L_{\text{beam}} \sim N(2900, \sigma^2_{\text{beam}})$ the distance between beams, independent of L. Define the gap

$$D = L - L_{\text{beam}}.$$

The requirement is $90 < D < 110$, and we want this to hold with probability 0.99.

Step 2: Find the mean and variance of D using Theorem 2.56 (mean and variance of linear combinations).

D is the linear combination $D = 1 \cdot L + (-1) \cdot L_{\text{beam}}$, so Theorem 2.56 gives:

$$E(D) = E(L) - E(L_{\text{beam}}) = 3000 - 2900 = 100,$$

$$V(D) = 1^2\, V(L) + (-1)^2\, V(L_{\text{beam}}) = 9 + \sigma^2_{\text{beam}}.$$

Step 3: Recognize that D is itself normally distributed.

A linear combination of independent normal random variables is again normal, so

$$D \sim N\left(100,\ 9 + \sigma^2_{\text{beam}}\right).$$

Step 4: Rewrite the requirement as a symmetric interval around the mean.

Since 90 and 110 are both exactly 10 away from $E(D) = 100$, the requirement $90 < D < 110$ is the same as $|D - 100| < 10$, i.e. a symmetric interval around the mean. So we need

$$P(90 < D < 110) = P(-10 < D - 100 < 10) = 0.99.$$

Step 5: Standardize using Theorem 2.43 (transformation to the standardized normal).

Let $\sigma_D = \sqrt{9 + \sigma^2_{\text{beam}}}$ and $Z = (D - 100)/\sigma_D$. Dividing the inequality by σ_D:

$$P\left(-\frac{10}{\sigma_D} < Z < \frac{10}{\sigma_D}\right) = 0.99.$$

Step 6: Match this to the given quantile $z_{0.995}$.

For a standard normal, a symmetric interval $(-c, c)$ that captures probability 0.99 leaves 0.005 in each tail, so c is exactly the 99.5% quantile: $c = z_{0.995} = 2.576$. Therefore

$$\frac{10}{\sigma_D} = 2.576.$$

Step 7: Solve for σ_D, then for σ_{beam}.

$$\sigma_D = \frac{10}{2.576}, \qquad \sigma_D^2 = 9 + \sigma_{\text{beam}}^2 \implies \sigma_{\text{beam}}^2 = \left(\frac{10}{2.576}\right)^2 - 9.$$

Step 8: Compute the numerical answer.

$$\sigma_{\text{beam}} = \sqrt{\left(\frac{10}{2.576}\right)^2 - 9} \approx 2.464 \text{ mm}.$$

To conclude we find that σ_{beam} must be no larger than 2.464 mm: $\sigma_{\text{beam}} \leq 2.464$ mm.

Exam-style questions

c) A bolt manufacturer requires the diameter $D \sim N(10, 0.05^2)$ mm to satisfy 9.90 mm $\leq D \leq$ 10.08 mm. Note that this interval is *not* symmetric about the mean 10. What is the correct way to compute the expected proportion of bolts falling outside this specification?

1) $P(D < 9.90) + P(D > 10.08)$, with each tail computed separately.
2) $2 \times P(D < 9.90)$, since the mean lies inside the interval.
3) $2 \times P(D > 10.08)$, since defects are more likely on the high side.
4) $P(D < 9.90) \times P(D > 10.08)$
5) $P(D < 9.90) - P(D > 10.08)$

|||| Solution

Correct answer: **1)**. Doubling one tail is only valid when the interval is symmetric about its mean, as in the question above. Here $10.08 - 10 = 0.08 \neq 10 - 9.90 = 0.10$, so the two tails must be found separately and added:

```
print(stats.norm.cdf(9.90, 10, 0.05) + 1 - stats.norm.cdf
    (10.08, 10, 0.05))
0.0775
```

d) Window panes have height $H \sim N(1200, \sigma_H^2)$ mm and are installed above door frames with mean height $\mu_{\text{frame}} = 1150$ mm and standard deviation $\sigma_{\text{frame}} = 8$ mm. The installation gap must satisfy $30 \text{ mm} < H - H_{\text{frame}} < 70$ mm in 95% of installations. What is the requirement for σ_H?
(Quantile from standard normal distribution: $z_{0.975} = 1.960$).

1) $\sigma_H \lessapprox 6.33$ mm
2) $\sigma_H \lessapprox 10.20$ mm
3) $\sigma_H \lessapprox 12.65$ mm
4) $\sigma_H \gtrapprox 40.12$ mm
5) $\sigma_H \gtrapprox 1.96$ mm

|||| Solution

Correct answer: **1)**. The interval $[30, 70]$ is symmetric about $\mathrm{E}(H - H_{\text{frame}}) = 1200 - 1150 = 50$, with half-width 20. We need:

$$z_{0.975} = \frac{20}{\sqrt{\sigma_H^2 + \sigma_{\text{frame}}^2}} = 1.960 \implies \sigma_H = \sqrt{\left(\frac{20}{1.960}\right)^2 - 8^2} = \sqrt{104.12 - 64} = 6.33\,\text{mm}.$$

2.13 Online statistics video views

|||| Exercise 2.13 Online statistics video views

In 2013, there were 110,000 views of the DTU statistics videos available online. Assume that the occurrence of views through 2014 follows a Poisson process with the 2013 average: $\lambda_{365\text{days}} = 110{,}000$.

a) What is the probability that in a randomly chosen half-hour period there are no views?

|||| Solution

The half-hour intensity is:

$$\lambda_{30\text{min}} = \frac{110{,}000}{365 \times 24 \times 2} = 6.28.$$

Let $X \sim Po(6.28)$:

$$P(X = 0) = e^{-6.28} = 0.00187.$$

```
lam = 110000 / (365*24*2)
print(stats.poisson.pmf(0, lam))
0.00187
```

b) There has just been a view. What is the probability of waiting more than 15 minutes for the next view?

Solution

The 15-minute rate is $\lambda_{15\text{min}} = 6.28/2 = 3.14$. The probability of zero views in 15 minutes is:

$$P(X = 0) = e^{-3.14} = 0.043.$$

Equivalently, using the exponential distribution with mean waiting time $\beta = 365 \times 24 \times 60/110{,}000 = 4.778$ minutes:

$$P(T > 15) = e^{-15/\beta} = e^{-3.14} = 0.043.$$

```
beta = 365*24*60/110000
print(1 - stats.expon.cdf(15, scale=beta))
```

Exam-style questions

c) A popular podcast received 84,000 downloads over the course of a 365-day year, modelled as a Poisson process. Which of the following Python expressions correctly computes the rate parameter (intensity) of downloads per 10-minute period?

1) `84000 * (365*24*6)`
2) `84000 / (365*24*60)`
3) `84000 / (365*24*6)`
4) `84000 / (365*6)`
5) `84000 / 365`

Solution

Correct answer: **3)**. There are $24 \times 6 = 144$ ten-minute periods per day (6 per hour), so $\lambda_{10\text{min}} = 84000/(365 \times 24 \times 6) \approx 1.598$.

```
print(84000 / (365*24*6))
1.598
```

2.14 Body mass index distribution

|||| Exercise 2.14 Body mass index distribution

The BMI (Body Mass Index) is defined as the weight (W) in kg divided by the squared height (H) in metres: $BMI = W/H^2$. Assume that the population distribution of BMI follows a log-normal distribution with $\alpha = 3.1$ and $\beta = 0.15$ (i.e. $\log(BMI) \sim N(3.1,\ 0.15^2)$).

a) A definition of "being obese" is a BMI value of at least 30. What proportion of the population would be obese?

|||| Solution

$$P(BMI > 30) = P(\log(BMI) > \log(30)) = P\left(Z > \frac{\log(30) - 3.1}{0.15}\right) = P(Z > 2.008) = 0.0223.$$

About 2.23% of the population would be obese.

```
print(1 - stats.lognorm.cdf(30, s=0.15, scale=np.exp(3.1)))
```

Exam-style questions

b) Income in a certain population follows a log-normal distribution with $\log(\text{Income}) \sim N(10.5, 0.4^2)$ (income measured in dollars). Which of the following Python expressions correctly computes the probability that a randomly selected person earns more than \$50,000?

1) `1 - stats.lognorm.cdf(50000, s=0.4, scale=np.exp(10.5))`
2) `1 - stats.lognorm.cdf(50000, s=10.5, scale=np.exp(0.4))`
3) `1 - stats.lognorm.cdf(50000, s=0.4, scale=10.5)`
4) `1 - stats.norm.cdf(50000, 10.5, 0.4)`
5) `1 - stats.lognorm.cdf(np.log(50000), s=0.4, scale=np.exp(10.5))`

Solution

Correct answer: **1)**. SciPy's `lognorm` expects the *shape* parameter `s`$= \sigma$ (here 0.4) and `scale`$= e^{\mu}$ (here $e^{10.5}$); it already applies the log-transform internally, so the raw value (50000, not its log) is passed in. Option 2) swaps μ and σ. Option 3) forgets to exponentiate μ for the scale parameter. Option 4) wrongly applies a plain normal distribution directly to the untransformed income value. Option 5) redundantly takes $\log(50000)$ *in addition to* using `lognorm`, effectively log-transforming twice.

```
print(1 - stats.lognorm.cdf(50000, s=0.4, scale=np.exp(10.5)
    ))
```

c) For the income distribution above ($\log(\text{Income}) \sim N(10.5, 0.4^2)$), which of the following statements is true?

1) The median income is $e^{10.5} \approx$ \$36,320, which is *less* than the mean income $e^{10.5+0.4^2/2} \approx$ \$39,340, because the log-normal distribution is right-skewed.
2) The median and mean income are both exactly $e^{10.5} \approx$ \$36,320, since $\log(\text{Income})$ is normally (and hence symmetrically) distributed.
3) The median income is 10.5 dollars, reading the parameter off directly.
4) The mean income is *less* than the median income, since log-normal distributions are left-skewed.
5) The mean and median cannot be determined without simulating the distribution.

Solution

Correct answer: **1)**. For $\log(X) \sim N(\mu, \sigma^2)$, the median of X is e^{μ}, but the mean is $e^{\mu+\sigma^2/2}$ — strictly larger than the median whenever $\sigma > 0$. This reflects the right skew of the log-normal distribution: symmetry of the underlying normal distribution for $\log(X)$ does *not* carry over to symmetry of X itself, since the exponential transform stretches the upper tail. Option 4) has the direction of skew backwards.

2.15 Bivariate normal

|||| Exercise 2.15 Bivariate normal

a) In the bivariate normal distribution, show that if Σ is a diagonal matrix then X_1 and X_2 are independent and each follows a univariate normal distribution.

|||| Solution

If $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$, then $|\Sigma| = \sigma_1^2\sigma_2^2$ and $\Sigma^{-1} = \text{diag}(1/\sigma_1^2, 1/\sigma_2^2)$. The joint density becomes:

$$\begin{aligned} f(x_1, x_2) &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\left(-\frac{(x_1-\mu_1)^2}{2\sigma_1^2} - \frac{(x_2-\mu_2)^2}{2\sigma_2^2}\right) \\ &= \underbrace{\frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-\frac{(x_1-\mu_1)^2}{2\sigma_1^2}}}_{f_{X_1}(x_1)} \cdot \underbrace{\frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-\frac{(x_2-\mu_2)^2}{2\sigma_2^2}}}_{f_{X_2}(x_2)}. \end{aligned}$$

Since the joint density factors into the product of two univariate normal densities, X_1 and X_2 are independent.

b) Let Z_1 and Z_2 be independent standard normal random variables. Define:

$$\begin{aligned} X &= a_{11}Z_1 + c_1, \\ Y &= a_{12}Z_1 + a_{22}Z_2 + c_2. \end{aligned}$$

Show that an appropriate choice of $a_{11}, a_{12}, a_{22}, c_1, c_2$ can give any bivariate normal distribution for (X, Y) with given $\boldsymbol{\mu}$ and covariance matrix Σ.

Solution

Matching means and covariances:

$$
\begin{aligned}
c_1 &= \mu_X, \quad c_2 = \mu_Y, \\
a_{11} &= \sqrt{\Sigma_{11}}, \\
a_{12} &= \frac{\Sigma_{12}}{\sqrt{\Sigma_{11}}}, \\
a_{22} &= \sqrt{\Sigma_{22} - \frac{\Sigma_{12}^2}{\Sigma_{11}}}.
\end{aligned}
$$

c) Use the result to simulate 1000 realisations of a bivariate normal random variable with $\boldsymbol{\mu} = (1, 2)^\top$ and

$$
\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},
$$

and make a scatter plot of the two components.

Solution

Since all questions involving simulation produce plots, Python code is omitted here; the simulation follows directly by substituting the values above into the formulas of the previous question.

With $\Sigma_{11} = 1$, $\Sigma_{12} = 1$, $\Sigma_{22} = 2$: $a_{11} = 1$, $a_{12} = 1$, $a_{22} = 1$, $c_1 = 1$, $c_2 = 2$. The scatter plot should show a positive correlation between X and Y.

Exam-style questions

d) Let (X, Y) follow a bivariate normal distribution with correlation $\rho = 0$ (i.e. $\mathrm{Cov}(X, Y) = 0$). Which of the following statements is correct?

1) X and Y must be independent — for *jointly normal* random variables, zero correlation implies independence, a special property that does not hold for arbitrary random variables.

2) X and Y are uncorrelated but not necessarily independent, since zero correlation never implies independence for any pair of random variables.

3) X and Y must be identically distributed, since $\rho = 0$ forces $\sigma_X = \sigma_Y$.

4) X and Y cannot both be normally distributed if $\rho = 0$.

5) There is not enough information to say anything about the relationship between X and Y.

|||| Solution

Correct answer: **1)**. This follows directly from the first question of this exercise: when Σ is diagonal (which is exactly what $\rho = 0$ means), the joint density factors into the product of the two marginal densities, which is the definition of independence. In general (for arbitrary, not necessarily normal, random variables) zero correlation only rules out a *linear* relationship and does *not* imply independence — but the bivariate/multivariate normal distribution is a special case where it does. Option 2) states the general-case fact but wrongly applies it here, ignoring the joint-normality assumption.

e) Using the same linear-transformation method ($X = a_{11}Z_1 + c_1$, $Y = a_{12}Z_1 + a_{22}Z_2 + c_2$, for independent standard normal Z_1, Z_2) as in the previous question, which choice of coefficients correctly simulates a bivariate normal with $\boldsymbol{\mu} = (0, 5)^\top$ and $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}$?

1) $a_{11} = 2, a_{12} = 1, a_{22} = 2, c_1 = 0, c_2 = 5$

2) $a_{11} = 4, a_{12} = 2, a_{22} = 5, c_1 = 0, c_2 = 5$

3) $a_{11} = 2, a_{12} = 2, a_{22} = 1, c_1 = 0, c_2 = 5$

4) $a_{11} = 2, a_{12} = 1, a_{22} = 2, c_1 = 5, c_2 = 0$

5) $a_{11} = \sqrt{2}, a_{12} = 1, a_{22} = 2, c_1 = 0, c_2 = 5$

Solution

Correct answer: **1)**. Using the formulas derived in the previous question:

$$
\begin{aligned}
a_{11} &= \sqrt{\Sigma_{11}} = \sqrt{4} = 2, \\
a_{12} &= \frac{\Sigma_{12}}{\sqrt{\Sigma_{11}}} = \frac{2}{2} = 1, \\
a_{22} &= \sqrt{\Sigma_{22} - \Sigma_{12}^2/\Sigma_{11}} = \sqrt{5 - 4/4} = \sqrt{4} = 2,
\end{aligned}
$$

with $c_1 = \mu_X = 0$ and $c_2 = \mu_Y = 5$. Option 2) mistakenly plugs the raw entries of Σ in directly, without applying the square-root/division formulas at all. Option 3) swaps a_{12} and a_{22}. Option 4) swaps the two means. Option 5) uses $\sqrt{\Sigma_{12}}$ in place of $\sqrt{\Sigma_{11}}$ for a_{11}.

2.16 Sample distributions

|||| Exercise 2.16 Sample distributions

a) Verify by simulation that

$$\frac{n_1 + n_2 - 2}{\sigma^2} S_p^2 \sim \chi^2(n_1 + n_2 - 2),$$

where S_p^2 is the pooled sample variance. You may use $n_1 = 5$, $n_2 = 8$, $\mu_1 = 2$, $\mu_2 = 4$, and $\sigma^2 = 2$.

|||| Solution

Simulate k independent pairs of samples, compute the pooled variance each time, scale by $(n_1 + n_2 - 2)/\sigma^2$, and compare the histogram to a $\chi^2(11)$ density. The two should match closely for large k. (Python code omitted as the exercise primarily produces diagnostic plots.)

b) Show that if $X \sim N(\mu_1, \sigma^2)$ and $Y \sim N(\mu_2, \sigma^2)$, then

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

Verify the result by simulation.

|||| Solution

From the previous question, $\frac{(n_1+n_2-2)S_p^2}{\sigma^2} \sim \chi^2(n_1 + n_2 - 2)$. Also,

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sigma \sqrt{1/n_1 + 1/n_2}} \sim N(0, 1).$$

Dividing the standard normal by the square root of the (scaled) chi-squared divided by its degrees of freedom yields a $t(n_1 + n_2 - 2)$ distribution by definition.

Simulation verification: compute the t-statistic for k simulated pairs of samples and compare the histogram to the $t(11)$ density. (Python code omitted as the exercise primarily produces diagnostic plots.)

Exam-style questions

c) Three independent samples of sizes $n_1 = 6$, $n_2 = 9$ and $n_3 = 4$ are drawn from populations sharing the same variance σ^2, and a pooled variance estimate S_p^2 is computed by pooling all three sample variances. What are the degrees of freedom of $\dfrac{n_1 + n_2 + n_3 - 3}{\sigma^2} S_p^2$?

1) 16
2) 19
3) 3
4) 13
5) 17

|||| Solution

Correct answer: **1)**. Each sample contributes $n_i - 1$ degrees of freedom (one is "used up" estimating that sample's own mean), so pooling three samples gives $(n_1 - 1) + (n_2 - 1) + (n_3 - 1) = n_1 + n_2 + n_3 - 3 = 6 + 9 + 4 - 3 = 16$ — the same principle as the two-sample case in the previous question of this exercise ($n_1 + n_2 - 2$), just generalised to three samples. Option 2) forgets to subtract anything. Option 3) subtracts a fixed 3 regardless of the sample sizes, rather than 1 per sample. Options 4) and 5) subtract the wrong constant (as if pooling 1 or 2 samples, respectively).

d) A statistic is constructed as $T = \dfrac{V}{\sqrt{W/m}}$, where $V \sim N(0,1)$ and $W \sim \chi^2(m)$. What is the distribution of T, and what key condition is required for this to hold?

1) $T \sim t(m)$, provided V and W are independent — this is precisely the definition of the t-distribution with m degrees of freedom.
2) $T \sim N(0,1)$, since dividing by $\sqrt{W/m}$ has no effect asymptotically.
3) $T \sim \chi^2(m)$, since W is chi-squared distributed.
4) $T \sim t(m)$ regardless of whether V and W are independent.
5) $T \sim F(1,m)$, since T is built from a normal and a chi-squared variable.

Solution

Correct answer: **1)**. This is exactly the mechanism used (implicitly) in the previous question: a standard normal divided by the square root of an independent chi-squared variable (divided by its degrees of freedom) follows a t-distribution with those degrees of freedom, by definition. Independence of V and W is essential — without it, the resulting distribution is generally not $t(m)$, so option 4) is false. Option 5) is a plausible-looking distractor: it is actually T^2 (not T itself) that would follow an $F(1, m)$ distribution.

2.17 Sample distributions 2

|||| Exercise 2.17 Sample distributions 2

Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ be independent, with $X_i \sim N(\mu_1, \sigma^2)$ and $Y_i \sim N(\mu_2, \sigma^2)$. Let S_1^2 and S_2^2 be the respective sample variances. Define:

$$Q = \frac{S_1^2}{S_2^2}.$$

a) For $n \in \{2, 4, 8, 16, 32\}$, find:

1. $P(Q < 1)$
2. $P(Q > 2)$
3. $P(Q < 1/2)$
4. $P(1/2 < Q < 2)$

|||| Solution

Q follows an F-distribution with degrees of freedom $\nu_1 = \nu_2 = n - 1$:

```
for n in [2, 4, 8, 16, 32]:
    F = stats.f(n-1, n-1)
    print(n, F.cdf(1), 1-F.cdf(2), F.cdf(0.5), F.cdf(2)-F.
       cdf(0.5))
```

As n increases, $P(Q < 1)$ stays near 0.5 (by symmetry of the F-distribution around 1), while $P(Q > 2)$ and $P(Q < 1/2)$ decrease (the ratio concentrates near 1).

b) For at least one value of n, verify the results above by direct simulation. You may use any values of μ_1, μ_2 and σ^2.

Solution

Simulate k pairs of samples of size n, compute $Q = S_1^2/S_2^2$ for each pair, and count how often each event occurs. The empirical proportions should match the theoretical F-distribution probabilities for large k. (Python code omitted as the exercise primarily validates via simulation plots.)

Exam-style questions

c) Two independent samples of sizes $n_1 = 10$ and $n_2 = 15$ are drawn from normal populations with equal variance σ^2. Let $Q = S_1^2/S_2^2$. What is the distribution of Q?

1) $Q \sim F(9, 14)$
2) $Q \sim F(10, 15)$
3) $Q \sim F(14, 9)$
4) $Q \sim t(23)$
5) $Q \sim \chi^2(23)$

Solution

Correct answer: **1)**. The ratio of two independent sample variances follows an F-distribution with degrees of freedom equal to $(n_1 - 1, n_2 - 1)$ for the numerator and denominator samples respectively: $Q = S_1^2/S_2^2 \sim F(9, 14)$. Option 2) uses the sample sizes directly instead of $n_i - 1$. Option 3) swaps the numerator and denominator degrees of freedom (these are *not* interchangeable, unlike for some symmetric distributions). Options 4) and 5) confuse Q with a t- or χ^2-statistic.

d) The previous exercise noted that $P(Q < 1)$ stays near 0.5 "by symmetry of the F-distribution around 1" when $n_1 = n_2 = n$ (so $\nu_1 = \nu_2 = n - 1$). If instead $n_1 = 10$ and $n_2 = 20$ (so $\nu_1 = 9 \neq \nu_2 = 19$), is $P(Q < 1)$ still exactly 0.5?

1) No — the F-distribution is only symmetric around 1 when $\nu_1 = \nu_2$; with unequal degrees of freedom, $P(Q < 1) \neq 0.5$ in general.

2) Yes — the F-distribution is always symmetric around 1, regardless of its degrees of freedom.
3) No — $P(Q < 1)$ becomes exactly 0 whenever $\nu_1 \neq \nu_2$.
4) Yes, but only because S_1^2 and S_2^2 are independent.
5) No — Q no longer follows an F-distribution at all once $n_1 \neq n_2$.

|||| Solution

Correct answer: **1)**. The exact symmetry $P(Q < 1) = P(Q > 1) = 0.5$ relied on $\nu_1 = \nu_2$ in the earlier calculation; it is not a general property of every F-distribution. With $\nu_1 = 9$ and $\nu_2 = 19$, the distribution of Q is skewed and $P(Q < 1)$ will differ from 0.5 (it remains true more generally that if $Q \sim F(\nu_1, \nu_2)$ then $1/Q \sim F(\nu_2, \nu_1)$, but this alone does not force $P(Q < 1) = 0.5$ unless $\nu_1 = \nu_2$). Q still follows an F-distribution regardless of whether $n_1 = n_2$ (ruling out option 5), just not a symmetric one.

```
print(stats.f(9, 19).cdf(1))
```

2.18 Stochastic variable

|||| Exercise 2.18 Stochastic variable

Let X be a stochastic variable with the following probability density function ($f(x)$):

X	0	1	2	3
$f(x)$	0.1	0.3	0.4	0.2

a) What is the sample space of X? Is X a discrete or a continuous stochastic variable?

|||| Solution

The sample space is $\{0, 1, 2, 3\}$. Since X can only take a finite, countable set of values (rather than any value in an interval), X is a *discrete* stochastic variable.

b) What is $P(X = 3)$ (the probability that X assumes the value 3)?

|||| Solution

Read directly from the table: $P(X = 3) = f(3) = 0.2$.

c) What is $P(X = 4)$ (the probability that X assumes the value 4)?

Solution

The value 4 is not part of the sample space of X, so $P(X = 4) = 0$.

d) What is the mean of X, $\mathrm{E}[X]$ (also called the expectation value)?

Solution

By Definition 2.13 (mean value),

$$\mathrm{E}[X] = \sum_x x \cdot f(x).$$

Applying this to the distribution of X:

$$\mathrm{E}[X] = \sum_x x \cdot f(x) = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 1.7.$$

e) What is the variance of X, $\mathrm{V}[X]$?
(This is question 3 from the May 2022 exam)

Solution

By Definition 2.16,

$$\mathrm{V}[X] = \mathrm{E}[(X - \mu)^2].$$

We compute:

$$\mathrm{V}[X] = (0 - 1.7)^2(0.1) + (1 - 1.7)^2(0.3) + (2 - 1.7)^2(0.4) + (3 - 1.7)^2(0.2) = 0.81,$$

Alternatively, use the fundamental identity $\mathrm{V}[X] = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$ (which is Eq. 4-5 in the textbook. If you have not encountered this identity before, it is a worthwhile exercise to prove it directly from the definition of variance).

$$\mathrm{E}[X^2] = 0^2(0.1) + 1^2(0.3) + 2^2(0.4) + 3^2(0.2) = 3.7,$$

$$\mathrm{V}[X] = 3.7 - 1.7^2 = 3.7 - 2.89 = 0.81.$$

Exam-style questions

f) A stochastic variable Y has sample space $\{1,2,3,4\}$ and probability density function $f(y)$, with $f(1) = 0.15$, $f(2) = g$, $f(3) = 0.25$, $f(4) = 0.10$, where g is unknown. What is g, and what is $P(Y \geq 3)$?

1) $g = 0.50$, $P(Y \geq 3) = 0.35$
2) $g = 0.50$, $P(Y \geq 3) = 0.10$
3) $g = 0.60$, $P(Y \geq 3) = 0.35$
4) $g = 0.50$, $P(Y \geq 3) = 0.85$
5) $g = 0.40$, $P(Y \geq 3) = 0.35$

|||| Solution

Correct answer: **1)**.
Since probabilities must sum to 1: $0.15 + g + 0.25 + 0.10 = 1 \Rightarrow g = 0.50$.
Also we compute: $P(Y \geq 3) = f(3) + f(4) = 0.25 + 0.10 = 0.35$.

g) Consider a stochastic variable Z with sample space $\{0,1,2\}$ and probability density function $f(z)$, with $f(0) = 0.2$, $f(1) = 0.5$, $f(2) = 0.3$. The following code was run in Python:

```
z = np.array([0, 1, 2])
fz = np.array([0.2, 0.5, 0.3])
print(np.sum(z * fz))

1.1

print(np.sum(z**2 * fz))

1.7
```

What is $\mathrm{V}[Z]$?

1) 0.49
2) 1.7
3) 0.6
4) 1.21
5) 2.91

|||| Solution

Correct answer: **1)**. The two printed values are $\mathrm{E}[Z] = 1.1$ and $\mathrm{E}[Z^2] = 1.7$, so — using the same $\mathrm{V}[Z] = \mathrm{E}[Z^2] - (\mathrm{E}[Z])^2$ formula as in the variance question above —

$$\mathrm{V}[Z] = 1.7 - 1.1^2 = 1.7 - 1.21 = 0.49.$$

Note that it is also possible to solve this questions by using Definition 2.16,

$$\mathrm{V}[X] = \mathrm{E}[(X - \mu)^2].$$

We compute:

$$\mu = \mathrm{E}[X] = 0(0.2) + 1(0.5) + 2(0.3) = 1.1,$$
$$\mathrm{V}[X] = (0 - 1.1)^2(0.2) + (1 - 1.1)^2(0.5) + (2 - 1.1)^2(0.3) = 0.49,$$